

WEEK 3 E25DA048

The screenshot shows the Power BI Desktop interface with the 'Data View' selected. The table 'SHOP_2015-05-21_...' contains the following columns: date, open, high, low, close, adj_close, volume, and date - Copy. The data spans from May 21, 2015, to June 16, 2015, with daily entries. The 'adj_close' column shows values like 2.75, 2.75, 2.75, etc. The 'volume' column shows values like 123039000, 28412000, 8202000, etc. The 'date - Copy' column shows dates like 21 May 2015, 22 May 2015, 26 May 2015, etc.

date	open	high	low	close	adj_close	volume	date - Copy
21-05-2015 00:00:00	2.7999995231628	2.87400007247925	2.41100001335144	2.56800007820129	2.56800007820129	123039000	21 May 2015
22-05-2015 00:00:00	2.607000011253357	3.10999998509583	2.59999994632557	2.831000008964539	2.831000008964539	28412000	22 May 2015
26-05-2015 00:00:00	2.98000001907349	3.03399991989136	2.90799999237061	2.96499991416931	2.96499991416931	8202000	26 May 2015
27-05-2015 00:00:00	3.0669999126196	3.081000008964539	2.70000004768372	2.75	2.75	7976000	27 May 2015
28-05-2015 00:00:00	2.75500011444092	2.7739999294281	2.64800000190735	2.744999988555908	2.744999988555908	4000000	28 May 2015
29-05-2015 00:00:00	2.72499990463257	2.73799991607666	2.68700003623962	2.72000002861023	2.72000002861023	2554000	29 May 2015
01-06-2015 00:00:00	2.72000002861023	2.74399995803833	2.66000008583069	2.73300004005432	2.73300004005432	2569000	01 June 2015
02-06-2015 00:00:00	2.74699997901917	2.74699997901917	2.66499996185303	2.6949999332428	2.6949999332428	3798000	02 June 2015
03-06-2015 00:00:00	2.67600011825562	2.75	2.65000009536743	2.74399995803833	2.74399995803833	3024000	03 June 2015
04-06-2015 00:00:00	2.74200010299683	2.74900007247925	2.55100011825562	2.67799997329712	2.67799997329712	3226000	04 June 2015
05-06-2015 00:00:00	2.66599988937378	2.73900008201599	2.60999989505583	2.73900008201599	2.73900008201599	1294000	05 June 2015
08-06-2015 00:00:00	2.69000005722046	2.74799990653992	2.68199992179871	2.69799995422363	2.69799995422363	2809000	08 June 2015
09-06-2015 00:00:00	2.66400003433228	2.84599995613098	2.66400003433228	2.83400011062622	2.83400011062622	3375000	09 June 2015
10-06-2015 00:00:00	2.84500002861023	2.88800001144409	2.79500007629395	2.80999994277954	2.80999994277954	2734000	10 June 2015
11-06-2015 00:00:00	2.84899997711182	3.07299995422363	2.84899997711182	3.07299995422363	3.07299995422363	4606000	11 June 2015
12-06-2015 00:00:00	3.1319999648242	3.36999988555908	3.125	3.36299991607666	3.36299991607666	8474000	12 June 2015
15-06-2015 00:00:00	3.3970000743866	3.81200003623962	3.29200005531311	3.54999995231628	3.54999995231628	16943000	15 June 2015
16-06-2015 00:00:00	3.60599994659424	3.83789992332458	3.58400011062622	3.81100010871887	3.81100010871887	11095000	16 June 2015
17-06-2015 00:00:00	3.96300005912781	4.21299982070923	3.80100011825562	3.86299991607666	3.86299991607666	15952000	17 June 2015
18-06-2015 00:00:00	3.84200000672939	3.88899993896484	3.4326000213623	3.5550000667572	3.5550000667572	11482000	18 June 2015
19-06-2015 00:00:00	3.63299989700317	3.67400002479553	3.25500011444092	3.55200004577637	3.55200004577637	9797000	19 June 2015
22-06-2015 00:00:00	3.67600011825562	3.84999990463257	3.60899996757507	3.69400000572205	3.69400000572205	7922000	22 June 2015
23-06-2015 00:00:00	3.77500009536743	3.80999994277954	3.516999959594568	3.54299998283386	3.54299998283386	2939000	23 June 2015
24-06-2015 00:00:00	3.51200008392334	3.74289989471436	3.41400003433228	3.59599995613098	3.59599995613098	5644000	24 June 2015
25-06-2015 00:00:00	3.63400000629425	3.65000009536743	3.48000001907349	3.48399996757507	3.48399996757507	2082000	25 June 2015
26-06-2015 00:00:00	3.4869998855591	3.53200006484985	3.26300001144409	3.3519999809265	3.3519999809265	3465000	26 June 2015
29-06-2015 00:00:00	3.25200009346008	3.3510000705719	3.1999988555908	3.13199996948242	3.13199996948242	3155000	29 June 2015

Stock Data

The Shopify stock dataset was imported into Power BI and examined in **Data View**. The dataset contains key stock-market attributes, including **Date**, **Open**, **High**, **Low**, **Close**, **Adjusted Close**, and **Volume**. These fields provide the necessary data for analysing stock-price trends, daily movements, and trading activity over time.

DATE TABLE:

The screenshot shows the Power BI Desktop interface with the 'Data View' selected. The table 'Date Table' contains the following columns: Date, Year, Quarter, Month, Month Number, and Day. The data spans from May 21, 2015, to June 16, 2015, with daily entries. The 'Year' column shows 2015, the 'Quarter' column shows Q2, the 'Month' column shows May or Jun, the 'Month Number' column shows 5 or 6, and the 'Day' column shows the day of the month (e.g., 21, 22, 23, etc.).

Date	Year	Quarter	Month	Month Number	Day
21-05-2015 00:00:00	2015	Q2	May	5	21
22-05-2015 00:00:00	2015	Q2	May	5	22
23-05-2015 00:00:00	2015	Q2	May	5	23
24-05-2015 00:00:00	2015	Q2	May	5	24
25-05-2015 00:00:00	2015	Q2	May	5	25
26-05-2015 00:00:00	2015	Q2	May	5	26
27-05-2015 00:00:00	2015	Q2	May	5	27
28-05-2015 00:00:00	2015	Q2	May	5	28
29-05-2015 00:00:00	2015	Q2	May	5	29
30-05-2015 00:00:00	2015	Q2	May	5	30
31-05-2015 00:00:00	2015	Q2	May	5	31
01-06-2015 00:00:00	2015	Q2	Jun	6	1
02-06-2015 00:00:00	2015	Q2	Jun	6	2
03-06-2015 00:00:00	2015	Q2	Jun	6	3
04-06-2015 00:00:00	2015	Q2	Jun	6	4
05-06-2015 00:00:00	2015	Q2	Jun	6	5
06-06-2015 00:00:00	2015	Q2	Jun	6	6
07-06-2015 00:00:00	2015	Q2	Jun	6	7
08-06-2015 00:00:00	2015	Q2	Jun	6	8
09-06-2015 00:00:00	2015	Q2	Jun	6	9
10-06-2015 00:00:00	2015	Q2	Jun	6	10
11-06-2015 00:00:00	2015	Q2	Jun	6	11
12-06-2015 00:00:00	2015	Q2	Jun	6	12
13-06-2015 00:00:00	2015	Q2	Jun	6	13
14-06-2015 00:00:00	2015	Q2	Jun	6	14
15-06-2015 00:00:00	2015	Q2	Jun	6	15
16-06-2015 00:00:00	2015	Q2	Jun	6	16

A dedicated **Date Table** was created using the **CALENDAR()** DAX function. The table includes **Date**, **Year**, **Quarter**, **Month**, **Month Number**, and **Day** fields, which support structured time-based analysis.

A **Year → Quarter → Month → Day** hierarchy was created to enable drill-down analysis and allow stock performance to be examined at different levels of the time period.

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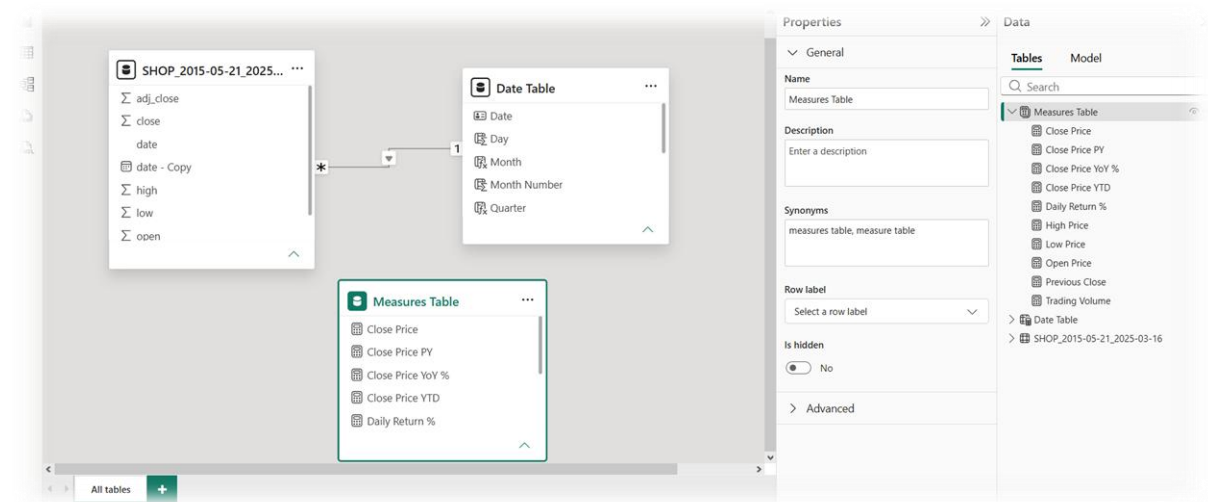
A **Year → Quarter → Month → Day** hierarchy was created to enable drill-down analysis and allow stock performance to be examined at different levels of the time period.

DAX Used

Date Table =

```
CALENDAR(  
    MIN('SHOP_2015-05-21_2025-03-16'[Date]),  
    MAX('SHOP_2015-05-21_2025-03-16'[Date])  
)
```

MEASURES TABLE:



A separate **Measures Table** was created to systematically organize the DAX measures used throughout the dashboard. The table contains financial measures including **Open Price, Close Price, High Price, Low Price, Trading Volume, and Daily Return %**.

It also includes time-intelligence measures such as **Previous Close, Close Price PY, Close Price YoY %, and Close Price YTD**, enabling year-over-year and year-to-date analysis.

The Measures Table remains **disconnected from the other tables**, as its purpose is solely to store and organize the measures used in the report.

DAX Measures

- **Close Price**

Close Price = AVERAGE('SHOP_2015-05-21_2025-03-16'[close])

- **Close Price PY**

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Close Price PY =

CALCULATE(
[Close Price],
SAMEPERIODLASTYEAR('Date Table'[Date])
)

- **Close Price YoY %**

Close Price YoY % =

DIVIDE(
[Close Price] - [Close Price PY],
[Close Price PY],
0
)

- **Close Price YTD**

Close Price YTD =

TOTALYTD(
[Close Price],
'Date Table'[Date]
)

- **Daily Return %**

Daily Return % =

DIVIDE(
[Close Price] - [Previous Close],
[Previous Close],
0
)

- **High Price**

High Price =

AVERAGE('SHOP_2015-05-21_2025-03-16'[high])

- **Low Price**

Low Price =

MIN('SHOP_2015-05-21_2025-03-16'[low])

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- **Open Price**

Open Price =

```
AVERAGE('SHOP_2015-05-21_2025-03-16'[open])
```

- **Previous Close**

Previous Close =

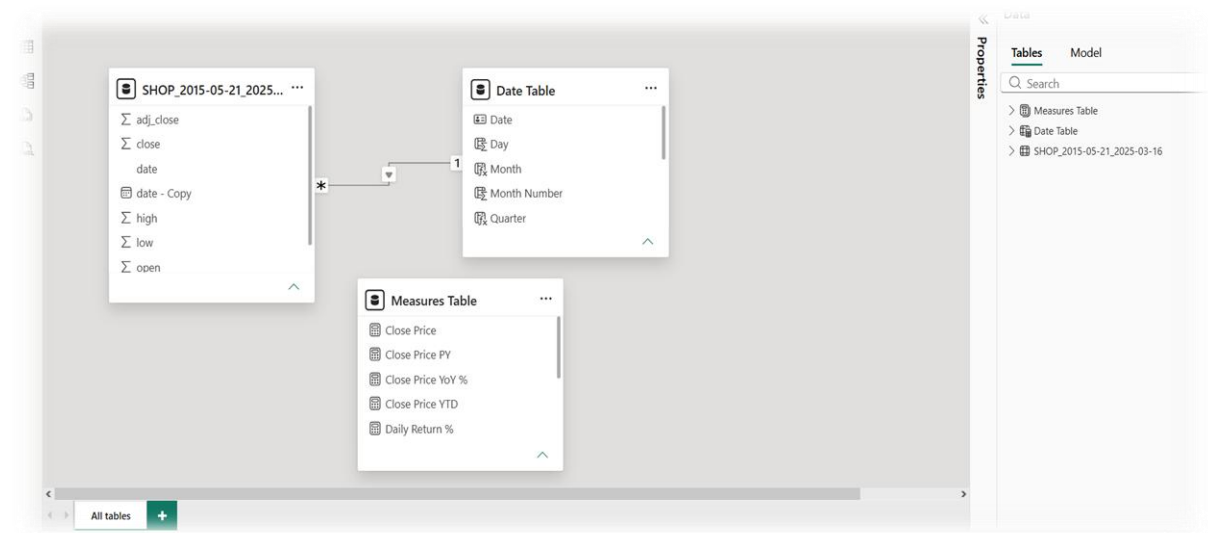
```
CALCULATE(  
    [Close Price],  
    DATEADD('Date Table'[Date], -1, DAY)  
)
```

- **Trading Volume**

Trading Volume =

```
SUM('SHOP_2015-05-21_2025-03-16'[volume])
```

MODEL VIEW

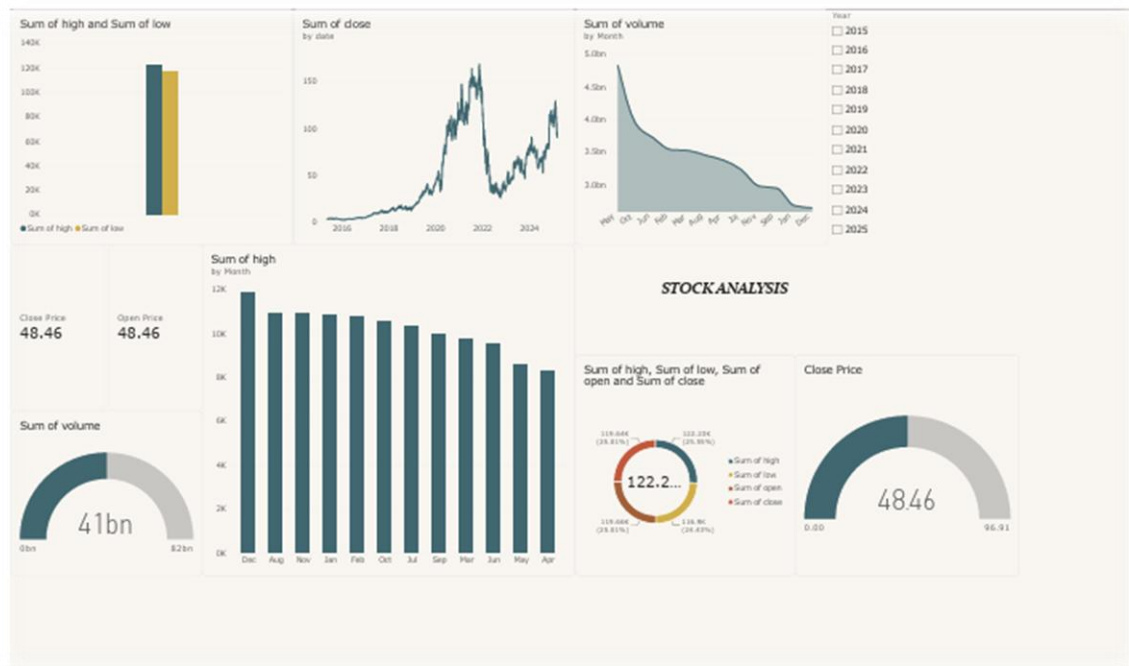


The **Model View** displays the relationship established between the Date Table and the Shopify stock data table. A **one-to-many relationship** was created between Date Table[Date] and the corresponding date field in the stock data table.

The relationship is **active** and uses **single-direction filtering**, allowing the Date Table to control the filtering of stock data during analysis.

The **Measures Table remains disconnected** from the data model because it serves only as a dedicated container for organizing DAX measures.

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An interactive **Stock Market Dashboard** was developed to provide a comprehensive view of Shopify's stock performance over time. The dashboard incorporates key visualizations covering **Close Price, Open Price and Close Price comparison, Close Price versus Previous Year, Daily Return %, Close Price YoY %, Trading Volume, and Close Price YTD**.

The **Year → Quarter → Month → Day** hierarchy was incorporated into the dashboard to support interactive drill-down and enable analysis across different time levels.

DRILL DOWN ANALYSIS:

The **Close Price** visualization supports hierarchical drill-down from **Year → Quarter → Month → Day**. This functionality allows users to move from a broader yearly view to increasingly detailed time periods and examine changes in Shopify's stock-price movement at each level.